Phase 1 · the ideas

Multilevel modeling

Three parts that build on each other.

Before we start

One running example, three parts

Part 1 · Why grouped data need special care

Part 2 · The random-intercept model: the ICC, centering, building and checking

Part 3 · Random slopes, a short extension

Part 1

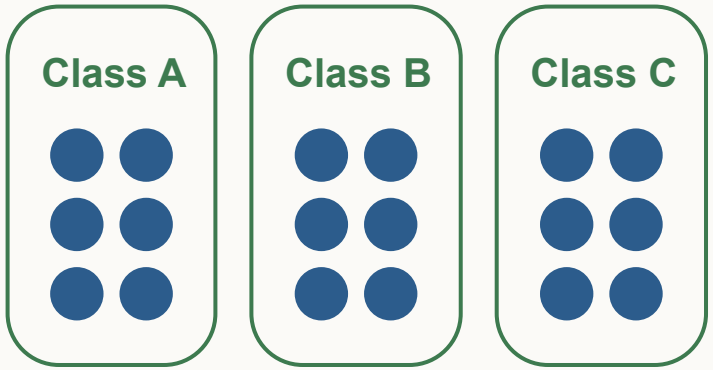
Why grouped data need special care

What changes when data come in groups?

Nested data: each student belongs to one class

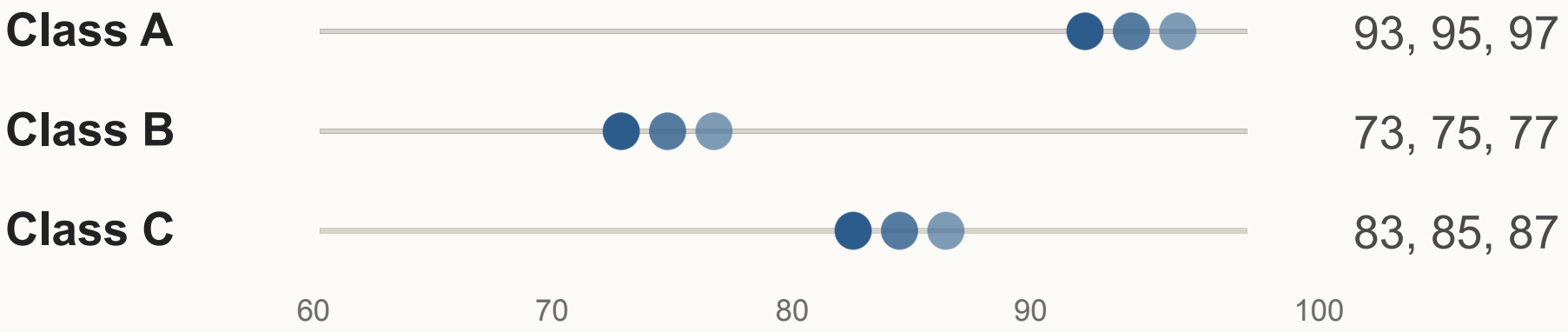
A school. Three math classes: A, B, C.

- student = level 1
  - class = level 2



Classmates score alike

Math test score (0–100)



Problem 1: regression counts too much evidence

What regression counts



9

9 students = 9 independent scores

What is really there



≈ 3

3 classmates score almost the same → about 3 scores

> Counting 3 as 9 makes regression overconfident → too many results called “significant”.

Problem 2: some questions are about the class

A question about a student

“Does a student who does more homework score higher?”

Ordinary regression: ✓ can answer

A question about a class

“Do smaller classes score higher?”

Ordinary regression: ✗ no class level

Multilevel model: ✓ has a class level

> “Class size” describes the whole class, not any one student. It needs its own level.

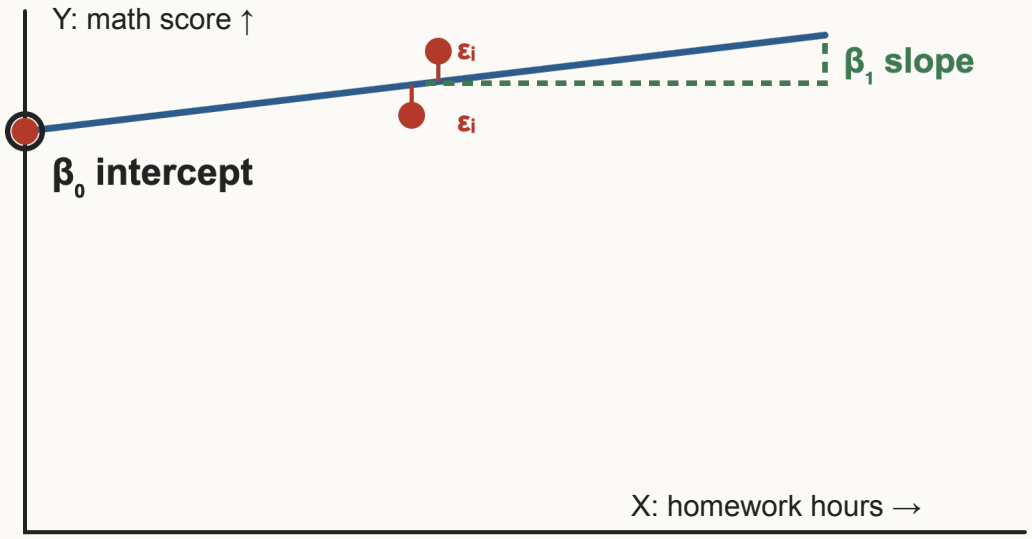
Part 2.1

The random-intercept model (Same Slope)

The core model.

Question: do students who do more homework score higher in math?

One class, one line



$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

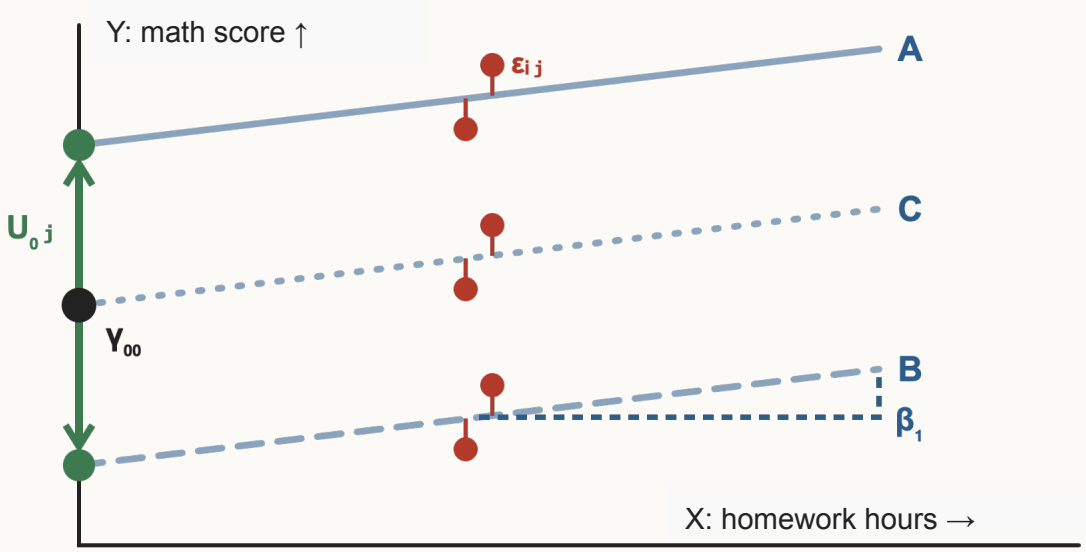
β_0 intercept
the score when x = 0

β_1 slope
score gain for 1 more hour of homework

ε_i residual = actual − line
how far each student is from the line. The best line makes these as small as possible.

Question: do students who do more homework score higher in math?

Three classes, three lines: four pieces



regression: $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$

$$y_{i\,j} = \gamma_{00} + \beta_1 x_{i\,j} + (U_{0\,j} + \varepsilon_{i\,j})$$

γ_{00} · typical intercept, the black dot

β_1 · slope, the same in every class

$U_{0\,j}$ · how far each class's line sits above or below the typical line

$\varepsilon_{i\,j}$ · how far each student sits from her own class's line

Multilevel model: the software uses maximum likelihood. Intuitively, it looks for the slope that keeps each student as close as possible to her own class line, while each class line is allowed to sit higher or lower.

Part 2.2

The ICC

Intraclass Correlation Coefficient

Where does the difference come from: the class, or the student?

$$y_{ij} = \gamma_{00} + \beta_1 x_{ij} + (U_{0j} + \varepsilon_{ij})$$

U_{0j}

From the class

Students score differently because they sit in different classes.
If the green gaps are large, the class matters a lot.

or

ε_{ij}

From the student

Students score differently because they are different people.
If the red gaps are large, the student matters a lot.

τ^2 = average of (class average – overall average)^2

σ^2 = average of (score – class average)^2

$$\mathrm{ICC} = \tau^2 \div (\tau^2 + \sigma^2)$$

ICC = 0 classes are all alike → grouping does not matter.
ICC = 1 classmates are all alike → one score per class.

"Spread" is called a variance in statistics: the average squared gap. Finch et al. (2024), eq. 2.1.

Step 1 · the two gaps, squared, then σ² and τ²

Class	Scores	Class avg	Score − class avg	Squared	Class avg − 85	Squared
A	93, 95, 97	95	−2, 0, +2	4, 0, 4	95 − 85 = +10	100
B	73, 75, 77	75	−2, 0, +2	4, 0, 4	75 − 85 = −10	100
C	83, 85, 87	85	−2, 0, +2	4, 0, 4	85 − 85 = 0	0

τ² · spread of the green gaps

$$\tau^2 = \text{sum of } (\text{class avg} - 85)^2 \div 3$$
$$= (100 + 100 + 0) \div 3$$
$$= 200 \div 3$$
$$\approx \mathbf{66.7}$$

σ² · spread of the red gaps

$$\sigma^2 = \text{sum of } (\text{score} - \text{class avg})^2 \div 9$$
$$= (4 + 0 + 4 + 4 + 0 + 4 + 4 + 0 + 4) \div 9$$
$$= 24 \div 9$$
$$\approx \mathbf{2.7}$$

Step 2: put σ² and τ² into the formula, then read the number.

Step 2 · the ICC, and what it means

Plug in

$$\text{ICC} = \tau^2 \div (\tau^2 + \sigma^2)$$
$$= 66.7 \div (66.7 + 2.7)$$
$$= 66.7 \div 69.4$$
$$\approx \mathbf{0.96}$$

What 0.96 means

96% of the difference in scores lies between classes; only 4% lies between classmates.

Knowing a student's class tells you almost everything about her score; who she is adds little.

The nine scores behave almost like three.



96% between classes **4% within classes**

Made-up numbers, extreme on purpose: in real education data the ICC is usually between 0.05 and 0.25.

Part 2

Building the multilevel model

The model we build

$$y_{ij} = \gamma_{00} + \beta_1 x_{ij} + U_{0j} + \varepsilon_{ij}$$

x_{ij} = student i's homework hours; y_{ij} = math score

γ_{00}

typical intercept

height of the typical line at x = 0 hours

β_1

slope

one more hour → how many more points

U_{0j}

class gap

how far class j's line sits from the typical line

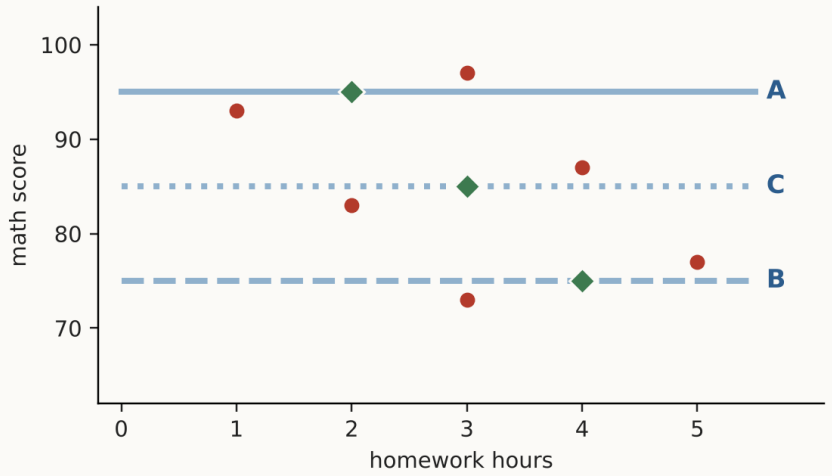
ε_{ij}

student gap

how far student i sits from her class's line

Model 0 (null / empty model): nothing added, just class averages

$y_{ij} = \gamma_{00} + U_{0j} + \varepsilon_{ij}$ with our numbers: $y_{ij} = 85$



Class	Scores	Flat line (class avg)	ε_{ij} = score − line	ε_{ij}^2	U_{0j} = line − 85
A	93, 95, 97	95	−2, 0, +2	4, 0, 4	+10
B	73, 75, 77	75	−2, 0, +2	4, 0, 4	−10
C	83, 85, 87	85	−2, 0, +2	4, 0, 4	0

Students: $\Sigma\, \varepsilon_{ij}^2 = 24 \rightarrow \sigma^2 = 24 \div 9 \approx 2.7$

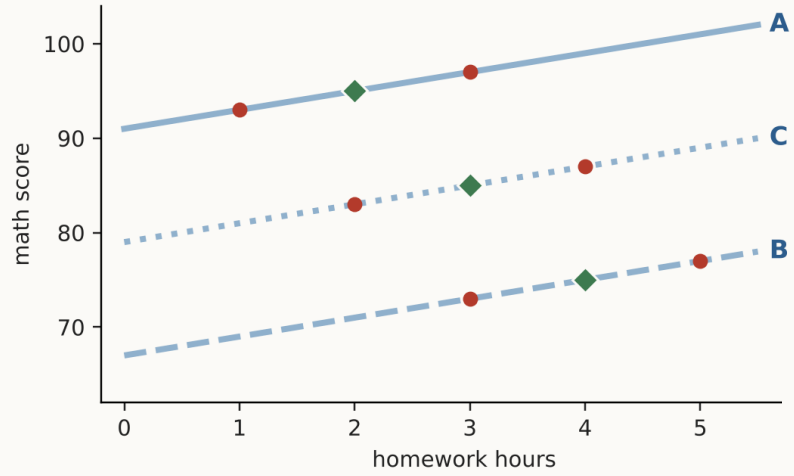
Classes: $\Sigma\, U_{0j}^2 = 100 + 100 + 0 = 200 \rightarrow \tau^2 = 200 \div 3 \approx 66.7$

ICC = $66.7 \div (66.7 + 2.7) \approx 0.96$

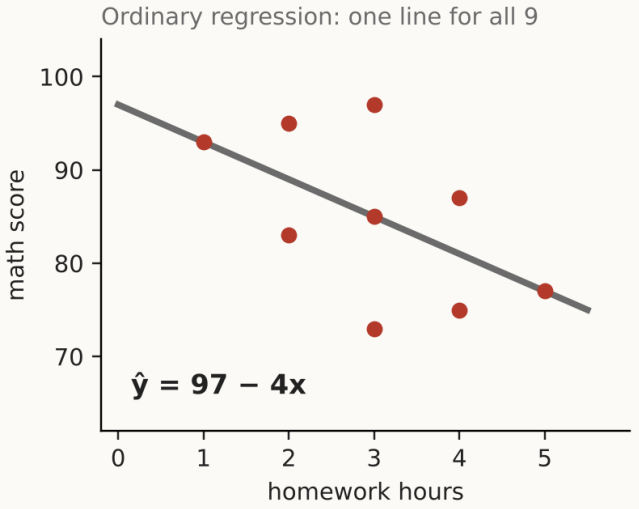
Model 1: add the student's homework hours

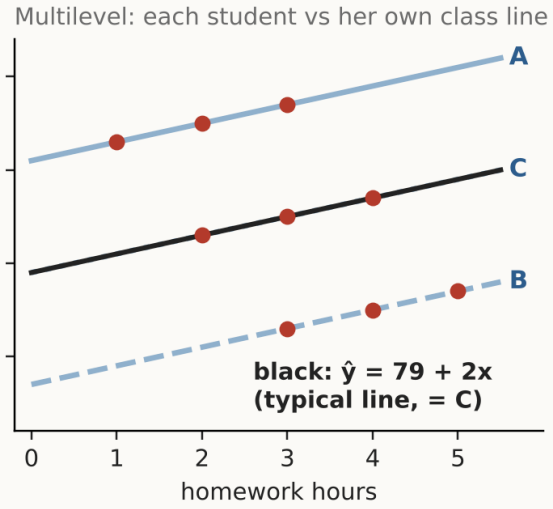
$$y_{ij} = \gamma_{00} + \beta_1 x_{ij} + U_{0j} + \varepsilon_{ij}$$

with our numbers: $y_{ij} = 79 + 2\,x_{ij}$



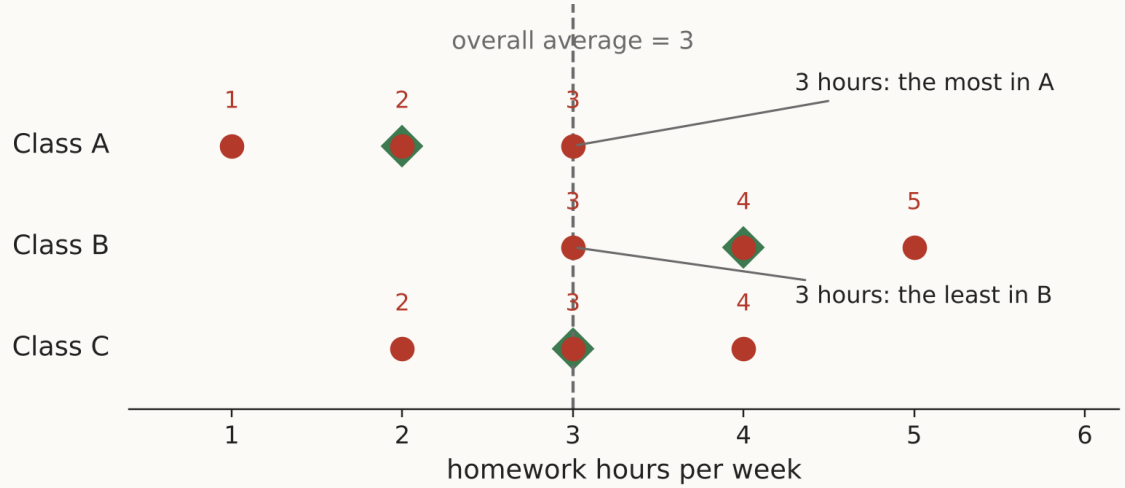
Class	Scores	Hours	Line: 2 × hours + ?	Line at 0 h
A	93, 95, 97	1, 2, 3	91 + 2 × hours	91
B	73, 75, 77	3, 4, 5	67 + 2 × hours	67
C	83, 85, 87	2, 3, 4	79 + 2 × hours	79





Two problem with current model

$$y_{ij} = \gamma_{00} + \beta_1 x_{ij} + U_{0j} + \varepsilon_{ij}$$



Class	Homework hours	Class average
A	1, 2, 3	2
B	3, 4, 5	4
C	2, 3, 4	3

Overall average: 3 hours

Two students, both at 3 hours

In A: $3 = 2 + 1$ → 1 above her classmates

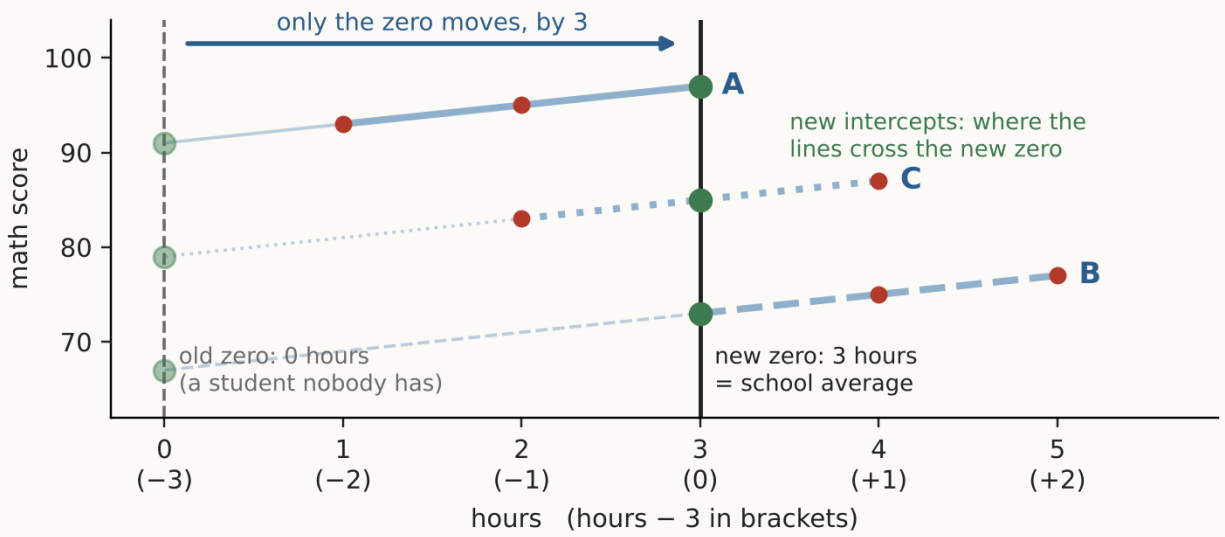
In B: $3 = 4 - 1$ → 1 below her classmates

> The same number means opposite things.

Part 2

Two ways of centering

Method 1 · Grand-mean centering: a new zero for everyone

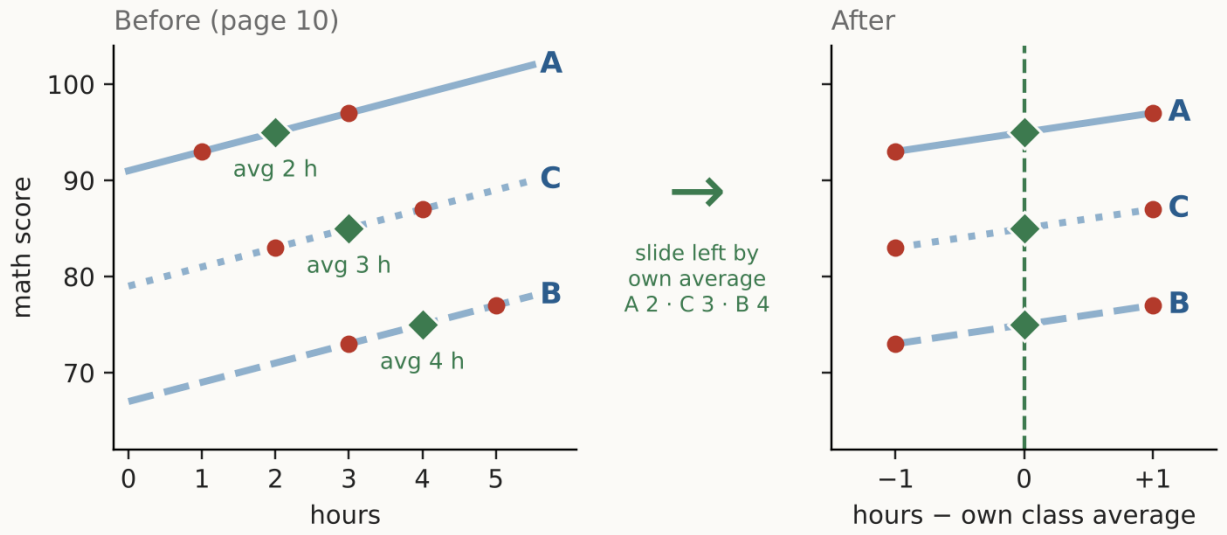


Class	Hours	Hours − 3
A	1, 2, 3	−2, −1, 0
B	3, 4, 5	0, +1, +2
C	2, 3, 4	−1, 0, +1

① **The intercept gets a real meaning.** The intercept is where a line crosses the zero line. Old zero line: a student with 0 hours, who does not exist (faint dots). New zero line: a student with average hours (dark dots).

$$y_{ij} = \gamma_{00} + \beta_1 (x_{ij} - 3) + U_{0j} + \varepsilon_{ij} \quad \text{with our numbers: } y_{ij} = 85 + 2 (x_{ij} - 3)$$

Method 2 · Group-mean centering: each class gets its own zero



Class	Hours	Hours − own class avg
A (avg 2)	1, 2, 3	1−2, 2−2, 3−2 = −1, 0, +1
B (avg 4)	3, 4, 5	3−4, 4−4, 5−4 = −1, 0, +1
C (avg 3)	2, 3, 4	2−3, 3−3, 4−3 = −1, 0, +1

① Each line slides left by its own class average
② The two 3-hour students are now different.
③ The new intercepts are the class averages: 95, 85, 75.

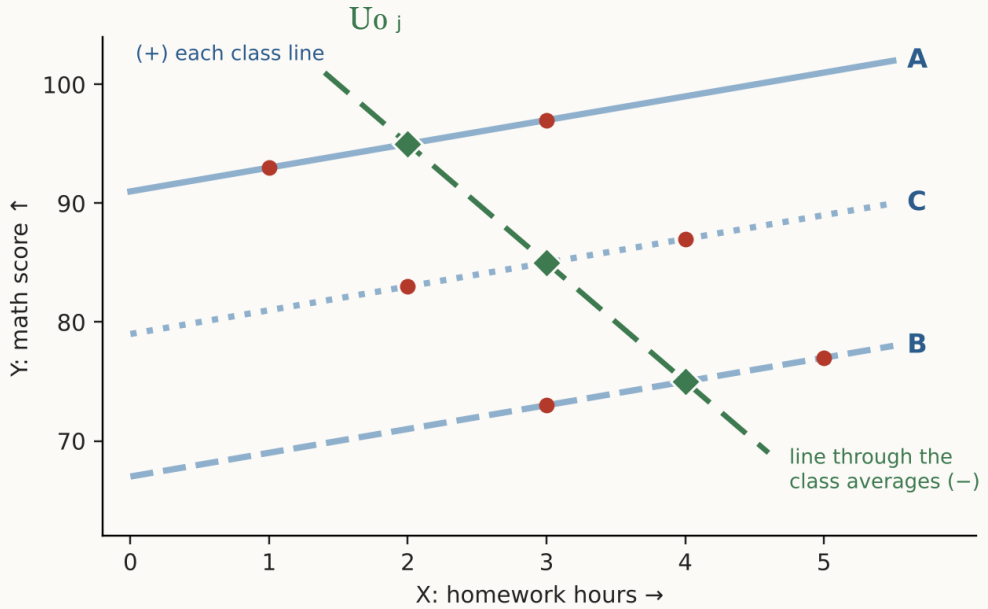
$$y_{ij} = \gamma_{00} + \beta_1 (x_{ij} - \bar{x}_j) + U_{0j} + \varepsilon_{ij}$$ with our data $\hat{y} = 85 + 2 (x_{ij} - \bar{x}_j)$

ε_{ij} But something is lost: the lines no longer sit at different hours. "B2 does 2 more hours than A2" has disappeared. Next page puts it back.

The class averages (2, 4, 3) were subtracted away on the last page. Put them back as a second column: one number per class, at the class level.

Method 2, step 2 · Put the class average back

Class	Hours	Class avg	Column 1 hours − class avg → level-1 x	Column 2 class avg → level-2 x
A	1, 2, 3	2	−1, 0, +1	2
B	3, 4, 5	4	−1, 0, +1	4
C	2, 3, 4	3	−1, 0, +1	3



Blue lines = column 1. Inside each class, more hours → higher score.
Slope = 2, the same in every class.
Green dashed line = column 2. It joins the three class averages (diamonds).

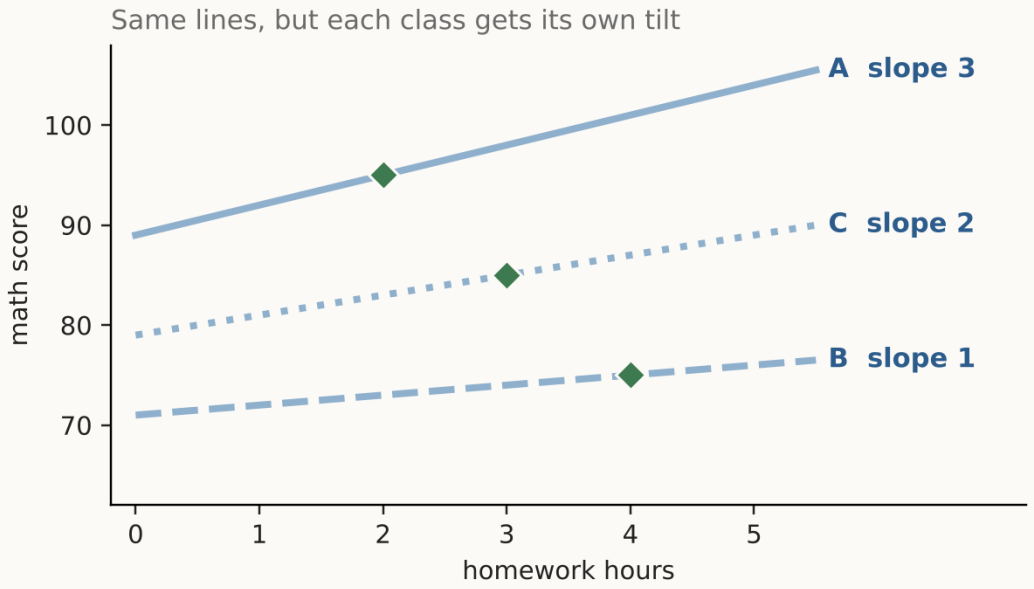
$$y_{ij} = \gamma_{00} + \beta_1 (x_{ij} - \bar{x}_j) + \gamma_{01} \bar{x}_j + U_{0j} + \varepsilon_{ij}$$

with our numbers: $y_{ij} = 115 + 2 (x_{ij} - \bar{x}_j) - 10 \bar{x}_j$

Part 3

Random slopes

Random slopes: each class gets its own tilt



When to use it

Theory says the effect of x differs across classes, and there are enough classes. Add it as one more piece; run the same deviance check.